
Welcome to Calculus I

Work these in class. Worked solutions are on the posted slide deck — each problem below names the slide to look for.

Day 1 • Problem 1*slide: Big Limit #1: How Fast, Right Now?*

Over an *interval* you can do it with algebra you already have. Using $s(t) = 16(-t^2 + 2t + 3)$, compute the average velocity $\frac{s(x) - s(a)}{x - a}$ with $a = 0.5$ — first for $x = 1$, then for $x = 0.6$.

Day 1 • Problem 2*slide: Do It Once with Letters*

Form $\frac{s(x) - s(0.5)}{x - 0.5}$ using $s(t) = 16(-t^2 + 2t + 3)$, and simplify it as far as it will go. Factor the top.

Day 1 • Problem 3*slide: Big Limit #2: Area by Exhaustion*

Find the exact area of the inscribed **square**, then the inscribed **regular hexagon**. Geometry only — no calculus, no calculator.

Day 1 · Problem 4 (worked together)

slide: Whose Example Is It?

I work it at the board — usually because it introduces a new idea, or because watching once beats being stuck for ten minutes.

Day 1 · Problem 5

slide: The Experiment You Are About to Run on Yourself

Predict: on that exam, how did the unrestricted-AI group do against the group that had no AI at all?